# Unit-3

# Simplex Algorithm

A factory makes **Gadgets ($x_1$)** and **Widgets ($x_2$)**.

- **Profit:** \$2 per Gadget, \$5 per Widget.
- **Labor Constraint:** $x_1 + x_2 \leq 10$ hours.
- **Machine Constraint:** $x_1 + 2x_2 \leq 12$ hours.

## Step 1: Convert to Standard Form

We add **Slack Variables** ($s_1$ and $s_2$) to turn the inequalities into equations.

1. $x_1 + x_2 + s_1 = 10$
2. $x_1 + 2x_2 + s_2 = 12$
3. **Objective rewritten for the table:** $Z - 2x_1 - 5x_2 - 0s_1 - 0s_2 = 0$

## Step 2: Create the Initial Tableau

| Basic | x1 | x2 | s1 | s2 | RHS |
|---|---|---|---|---|---|
| Z | -2 | **-5** | 0 | 0 | 0 |
| S1 | 1 | 1 | 1 | 0 | 10 |
| S2 | 1 | **2** | 0 | 1 | 12 |

**A. Pivot Column (The Entering Variable)**

This column tells you **which new product to start making** to increase your profit. It is chosen by looking for the most negative number in the bottom row (in a maximization problem).

**B. Pivot Row (The Leaving Variable)**

This row tells you **which resource will run out first**. You find it by doing the **Ratio Test** (dividing the RHS by the numbers in your pivot column). The smallest positive ratio "wins" the seat.

# Step 3: Identify the Pivot

**1. Find the Entering Variable (Pivot Column):** Look at the Z row. Which negative number is the "most negative"?

- **-5** is smaller than -2. So, **x2 (Widgets)** is our pivot column. We want to make Widgets because they give the most profit.

2. **Find the Leaving Variable (Pivot Row):** Divide the RHS by the numbers in the pivot column (Ratio Test).
   - Row $s_1$: $10 \div 1 = 10$
   - Row $s_2$: $12 \div 2 = \mathbf{6}$
   - **Smallest ratio is 6.** So, **Row** $s_2$ is our pivot row. $s_2$ will leave the "basic" group.

**3.The Pivot Element:** The number where they meet is **2**

# Step 4: Perform Row Operations

We need to make the Pivot Element **1** and everything else in its column **0**.

1. **New Pivot Row ($R_2$):** Divide the whole row by 2.
   - $[0.5, \mathbf{1}, 0, 0.5, 6]$
2. **Update Z-Row ($R_Z$):** We need to eliminate the -5. We do $R_Z + 5(\text{New } R_2)$.
   - Result: $[0.5, \mathbf{0}, 0, 2.5, 30]$
3. **Update $s_1$ Row ($R_1$):** We need to eliminate the 1. We do $R_1 - (\text{New } R_2)$.
   - Result: $[0.5, \mathbf{0}, 1, -0.5, 4]$

# Step 5: The Final Tableau

We check the Z row. Are there any negative numbers left? **No!** (0.5, 0, 0, 2.5 are all positive). This means we have reached the **Optimal Solution**.

| Basic | x1 | x2 | s1(Labor Time) | s2(Machine Time) | RHS |
|---|---|---|---|---|---|
| **Z(Profit)** | 0.5 | 0 | 0 | 2.5 | **30** |
| **S1(Laor)** | 0.5 | 0 | 1 | -0.5 | **4** |
| **X2(Widget)** | 0.5 | 1 | 0 | 0.5 | **6** |

In the first column, you see $s_1$ and $x_2$. Think of these as the only items currently in your "production bag."

- $x_2 = 6$: This tells the manager, "Produce **6 Widgets**."
- $s_1 = 4$: This is your **Slack**. It tells the manager, "You have **4 hours of Labor** sitting unused."
- **Where is $x_1$?** Since $x_1$ (Gadgets) is **not** in the left-hand column, its value is **zero**. The table is saying: "Don't make any gadgets; they aren't as profitable as widgets right now."

Read the **Basic Variables** and the **RHS**:

- **x2 = 6:** Produce **6 Widgets**.
- **x1 = 0:** Produce **0 Gadgets** (since x1 is not in the Basic column).
- **Z = 30$:** Maximum profit is **$30**.
- **s1 = 4:** You have **4 hours of Labor** leftover (Slack).
- **s2 = 0:** You have **0 hours of Machine time** leftover

## Summary of Simplex

- **Standardize** (Add Slacks).
- **Tableau** (Fill the grid).
- **Pivot** (Pick the best variable to increase).
- **Repeat** (Until no negatives remain in the Z row)

$s_2$

# (Second Row)

| Variable | Value | The "Real World" Meaning |
|---|---|---|
| S1 | **1** | This confirms this row is tracking your 1 unit of leftover Labor. |
| x1 | **0.5** | **The Consumption:** If you decided to make 1 Gadget (x1), it would "eat up" 0.5 hours of your remaining Labor slack. |
| x2 | **-0.5** | **The Helper :** The negative sign means if you get more Machine time (s2), it actually **reduces** your wasted labor by 0.5 hours because you can put that labor to work! |
| **RHS** | **4** | **The Waste:** This is exactly how many Labor hours are currently "wasted". |

# (Third Row)

| Variable | Value | Meaning |
|---|---|---|
| x2 | **1** | This is the "Identity." It just confirms this row represents 1 Widget. |
| x1 | **0.5** | **The Trade-off:** To make 1 extra Gadget (x1), you have to **sacrifice** 0.5 Widgets (x2) because they share resources. |
| S2 | **0.5** | **The Dependency:** Every 1 unit of Machine Time (s2) you "lose" or "give away" would cost you 0.5 Widgets in production. |
| **RHS** | **6** | This is how many Widgets you are currently holding in your hand. |

When you set up a Z-row, you are looking at the **Net Evaluation** of a variable.

- In Maximization: A negative coefficient in the Z-row means "This variable will **increase** profit."
- In Minimization: A positive coefficient in the Z-row means "This variable will **decrease** cost."

If you just pick the "least negative" (or most positive) and start pivoting, you might move toward a better cost, but you will be doing it with a **negative surplus variable** ($s_1 = -12$). The math will eventually crash because you can't have a "negative amount" of leftover resources.

The solid reasoning is that **Minimization is not the mirror image of Maximization** when it comes to the starting point of the math.

Here are the three fundamental reasons why simply changing the pivot rule isn't enough.

**The "Negative Inventory" Problem (Feasibility)**

In a **Maximization** problem with ≤ constraints, we add a Slack variable (+s). At the starting point (0,0), s=RHS. Since the RHS is positive, our "inventory" is positive. The math is happy.

In a **Minimization** problem with ≥ constraints, we must use a **Surplus variable** (−s).

- Equation: $2x1+3x2-s1=12$
- If we start at the origin (0,0), the equation becomes: $0+0-s1=12 \rightarrow s1=-12$.

**The Core Rule:** Simplex variables **must** be non-negative. A "Basic Variable" cannot be −12. If you try to start a table with a negative value in the RHS, the Ratio Test (which determines how far you can move) completely breaks down. You aren't just at a "bad" corner; you are **outside the shape entirely.**

The Problem is the "Origin" (0,0)
In the standard Simplex method, the math assumes you start at the origin: $x_1 = 0$ and $x_2 = 0$.

- **In Maximization :** (0,0) is a **legal** starting point. If you make 0 chairs and 0 tables, you use 0 wood and 0 labor. This doesn't break any rules.
- **In Minimization :** (0,0) is **illegal**. If the rule is "You must have at least 12 units of Protein," and you start with 0, you have violated the constraint.

**The Simplex math cannot function if your starting point is outside the "Legal Zone" (Feasible Region).**

If you try to use a standard tableau on a minimization problem with constraints without Big-M or Duality:

1. **The Slack is Negative:** Your equation would look like $2x_1 + 1x_2 + s_1 = 12$. If $x_1$ and $x_2$ are 0, then $s_1 = 12$. But in constraints, we use surplus: $2x_1 + 1x_2 - s_1 = 12$. If $x=0$, then **$s_1 = -12$**.
2. **Basic Variables must be Positive:** A core rule of Simplex is that your basic variables (the ones in the left column) **must be** .
3. **The "Broken" Table:** A table where $s_1 = -12$ is mathematically "broken." You cannot perform the Ratio Test accurately, and the algorithm won't know which way to walk to find the legal zone

# The Farm Feed Mix

A farmer needs to create a feed mix using two ingredients: **Grain ($x_1$)** and **Hay ($x_2$)**.

- **Goal:** Minimize the total cost.
- **Costs:** Grain costs **$3** per kg; Hay costs **$2** per kg.
- **Protein Requirement:** The mix must have at least **12 units** of protein.
  - Grain provides 2 units; Hay provides 1 unit.
- **Vitamin Requirement:** The mix must have at least **10 units** of vitamins.
  - Grain provides 1 unit; Hay provides 2 units.

**The Mathematical Model**

**Minimize:** $C = 3x_1 + 2x_2$

**Subject to:** 1. $2x_1 + 1x_2 \geq 12$ (Protein)

2. $1x_1 + 2x_2 \geq 10$ (Vitamins)

3. $x_1, x_2 \geq 0$

## Step 1: Create the Dual

**Maximize:** $W = 12y_1 + 10y_2$

**Subject to:** 1. $2y_1 + 1y_2 \leq 3$ (from the $x_1$ column)

2. $1y_1 + 2y_2 \leq 2$ (from the $x_2$ column)

Add slacks (S1, S2) to our Dual.

# Initial Tableau

| Basic | y1 | y2 | S1 | S2 | RHS |
|---|---|---|---|---|---|
| W | -12 | -10 | 0 | 0 | 0 |
| S1 | **2** | 1 | 1 | 0 | 3 |
| S2 | 1 | 2 | 0 | 1 | 2 |

1. **Entering:** $y_1$ (most negative, -12).
2. **Ratio Test:** $3/2 = \mathbf{1.5}$ vs $2/1 = 2$.
3. **Pivot Element: 2**.

## After First Pivot (Row Ops)

| Basic | y1 | y2 | S1 | S2 | RHS |
|---|---|---|---|---|---|
| W | 0 | -4 | 6 | 0 | 18 |
| y1 | 1 | 0.5 | 0.5 | 0 | 1.5 |
| S2 | 0 | **1.5** | -0.5 | 1 | 0.5 |

1. **Entering:** $y_2$ (-4).
2. **Ratio Test:** $1.5/0.5 = 3$ vs $0.5/1.5 = \mathbf{0.33}$.
3. **Pivot Element: 1.5**.

# The Final Tableau

| Basic | y1 | y2 | S1 | S2 | RHS |
|---|---|---|---|---|---|
| W | 0 | 0 | **4.67** | **2.67** | **19.33** |
| y1 | 1 | 0 | 0.67 | -0.33 | 1.33 |
| y2 | 0 | 1 | -0.33 | 0.67 | 0.33 |

1. **Min Cost ($C$):** \$19.33 (This is the RHS of the $W$ row).
2. **Amount of Grain ($x_1$):** Look under $S_1$. The value is **4.67 kg**.
3. **Amount of Hay ($x_2$):** Look under $S_2$. The value is **2.67 kg**.

- **Cost:** Liquid A costs **$5 per liter**; Liquid B costs **$8 per liter**.
- **Nitrogen Requirement:** The mix must contain at least **10 units** of Nitrogen.
    - Liquid A provides 2 units per liter.
    - Liquid B provides 1 unit per liter.
- **Phosphate Requirement:** The mix must contain at least **12 units** of Phosphate.
    - Liquid A provides 1 unit per liter.
    - Liquid B provides 3 units per liter.

- **Liquid A ($x_1$):** 3.6 liters
- **Liquid B ($x_2$):** 2.8 liters
- **Minimum Cost ($Z$): $40.40**

# Big-M Method

## Part 1: Formulate the Model

Write the Objective Function and the Constraints.

**Self-Check:** should have "Greater-than-or-equal-to" (≥) constraints.

## Part 2: Setup the Standard Form

- Add your **Surplus Variables** (s1,s2) by subtracting them.
- Add your **Artificial Variables** (A1,A2) to create a legal starting point.
- Assign the **Big-M** penalty to A1 and A2 in your objective function.

## Part 3: Build the Tableau

Place the coefficients into the grid. Remember to "clean" the Z-row so that A1 and A2 have zeros in their columns before you start pivoting.

**The Mathematical Model**

**Minimize:** $C = 3x_1 + 2x_2$

**Subject to:** 1. $2x_1 + 1x_2 \geq 12$ (Protein)

2. $1x_1 + 2x_2 \geq 10$ (Vitamins)

3. $x_1, x_2 \geq 0$

# Prepare the Standard Form

To turn into equations, we must subtract a **Surplus** and add an **Artificial Variable** and also add a massive penalty M to the objective function for any A used.

**Eq 1:** $2x_1 + 1x_2 - s_1 + A_1 = 12$

**Eq 2:** $1x_1 + 2x_2 - s_2 + A_2 = 10$

**Objective:** $Z = 3x_1 + 2x_2 + MA_1 + MA_2$

## The Initial Tableau

| **Basic** | **x1** | **x2** | **s1** | **s2** | **A1** | **A2** | **RHS** |
|---|---|---|---|---|---|---|---|
| Z | 3-3M | 2-3M | M | M | 0 | 0 | -22M |
| A1 | **2** | 1 | -1 | 0 | 1 | 0 | 12 |
| A2 | 1 | 2 | 0 | -1 | 0 | 1 | 10 |

**So, the thumb rule is:**

1. Prioritize the $M$ term.
2. If $M$ terms are equal, pick $x_1$ because it is the "first" variable. This prevents the algorithm from getting stuck in an infinite loop.

## Ratio Test

- For $x_1$: Ratios are $12/2 = \mathbf{6}$ and $10/1 = 10$.
- **Entering:** $x_1$ (Coefficient $3M - 3$)
- **Ratio Test:** $12/2 = \mathbf{6}$ (Row $A_1$) vs $10/1 = 10$ (Row $A_2$).
- **Pivot Element: 2**

**New Row 1 ($R_1 \div 2$):** $[1, 0.5, -0.5, 0, 0.5, 0, 6]$ **Update $Z$ and $A_2$:**

❖ Once an Artificial Variable leaves the "Basic" column, **it should never come back.**

- **Entering Variable:** $x_1$ came into the "Basic" column.
- **Leaving Variable:** $A_1$ was kicked out.

# After First Iteration

| Basic | x1 | x2 | s1 | s2 | A1 | A2 | RHS |
|---|---|---|---|---|---|---|---|
| Z | 0 | 1.5M-0.5 | 0.5M-1.5 | -M | …… | 0 | 4M+18 |
| X1 | 1 | 0.5 | -0.5 | 0 | 0.5 | 0 | 6 |
| A2 | 0 | **1.5** | 0.5 | -1 | -0.5 | 1 | 4 |

- **Entering:** $x_2$ (Coefficient $1.5M - 0.5$)
- **Ratio Test:** $6/0.5 = 12$ vs $4/1.5 = \mathbf{2.67}$.
- **Pivot Element: 1.5**

**New Row 2 ($R_2 \div 1.5$):** $[0, 1, 0.33, -0.67, -0.33, 0.67, 2.67]$

# The Final Tableau

After cleaning the $x_2$ column, the $Z$-row will no longer have $M$ terms (because $A_1$ and $A_2$ are gone).

| Basic | x1 | x2 | s1 | s2 | RHS |
|---|---|---|---|---|---|
| Z | 0 | 0 | 1.33 | 0.67 | **19.33** |
| X1 | 1 | 0 | -0.67 | 0.33 | **4.67** |
| x2 | 0 | 1 | 0.33 | -0.67 | **2.67** |

1. $x_1$ **(Grain): 4.67 kg**
2. $x_2$ **(Hay): 2.67 kg**
3. **Minimum Cost: $19.33**

**Row 1: The "Objective" Row ($Z$)**

This row represents your **Wallet**.

- **RHS = 19.33:** This is the **Minimum Cost**. It means that if you follow the instructions in the rows below, the cheapest you can feed your animals while meeting all requirements is **$19.33**.
- $x_1 = 0$ **and** $x_2 = 0$**:** This tells you the solution is **Stable**. Since there are no positive values here (in a minimization search), you cannot change $x_1$ or $x_2$ to lower the cost further.
- $s_1 = 1.33$ **and** $s_2 = 0.67$**:** These are the **Shadow Prices**. They tell you how much your cost would change if you were forced to increase your nutrient requirements by 1 unit.
  - If you needed 13 units of Protein (instead of 12), your cost would go up by **$1.33**.

**Row 2: The "$x_1$ (Grain)" Row**

This row represents the **Decision for Ingredient A**.

- **RHS = 4.67:** This is your **Purchase Order**. It tells you to buy exactly **4.67 kg of Grain**.
- $x_1 = 1$**:** This is just a mathematical confirmation that this row "belongs" to $x_1$.
- **The $s$ columns (-0.67 and 0.33):** These show the **Exchange Rate**. They tell the algorithm how much Grain ($x_1$) you would have to add or remove if you decided to change the amount of Hay ($x_2$) or change the requirements.

**Row 3: The "$x_2$ (Hay)" Row**

This row represents the **Decision for Ingredient B**.

- **RHS = 2.67:** This tells you to buy exactly **2.67 kg of Hay**.
- $x_2 = 1$**:** This confirms this row "belongs" to $x_2$.
- **The $s$ columns (0.33 and -0.67):** Similar to the row above, these values manage the balance between the two ingredients to ensure you don't drop below the minimum nutrient levels.

**1. The Problem: "The Daily Dose"**

A health lab needs to create a liquid supplement using two ingredients, **Liquid A ($x_1$)** and **Liquid B ($x_2$)**.

- **Minimize Cost:** $Z = 4x_1 + 1x_2$
- **Constraint 1 (Nutrient X):** $3x_1 + 1x_2 \geq 6$
- **Constraint 2 (Nutrient Y):** $4x_1 + 3x_2 \geq 12$
- **Non-negativity:** $x_1, x_2 \geq 0$

**Minimize the Cost ($Z$):** Nuts cost **$8** per kg, and Oatmeal costs **$5** per kg. **Protein** Must have at least **20 units**. (Nuts give 5 units/kg, Oatmeal gives 2 units/kg).

**Fat** Must not exceed **10 units**. (Nuts have 2 units/kg, Oatmeal has 1 unit/kg).

## Step 1: Convert to Standard Form

- **Constraint 1 ($\geq$):** Subtract Surplus ($s_1$) and add Artificial ($A_1$).
- **Constraint 2 ($\leq$):** Just add Slack ($s_2$).

1. $5x_1 + 2x_2 - s_1 + A_1 = 20$
2. $2x_1 + 1x_2 + s_2 = 10$
3. **Objective:** Minimize $Z = 8x_1 + 5x_2 + MA_1$

## Initial Tableau

| Basic | x1 | x2 | s1 | s2 | A1 | RHS |
|---|---|---|---|---|---|---|
| Z | 8-5M | 5-2M | M | 0 | 0 | -20M |
| A1 | **5** | 2 | -1 | 0 | 1 | 20 |
| S2 | 2 | 1 | 0 | 1 | 0 | 10 |

- **Entering Variable:** $x_1$ (Most positive $M$ term: $5M - 8$).
- **Ratio Test:** $20/5 = 4$ (Row $A_1$) vs $10/2 = 5$ (Row $s_2$).
- **Pivot Element: 5**.

# Updated Tableau

| Basic | x1 | x2 | s1 | s2 | A1 | RHS |
|---|---|---|---|---|---|---|
| Z | 0 | 0.2M-1.8 | 0.2M-1.6 | 0 | …… | 32 |
| x1 | 1 | 0.4 | -0.2 | 0 | 0.2 | 4 |
| s2 | 0 | 0.2 | 0.4 | 1 | -0.4 | 2 |

A1 has already left the basis. In Big-M, once A is gone, we usually treat M as **zero** or a very small number for the remaining "real" steps. Now, re-evaluate the Z-row without M.

| Basic | x1 | x2 | s1 | s2 | A1 | RHS |
|---|---|---|---|---|---|---|
| Z | 0 | -1.8 | -1.6 | 0 | …… | 32 |
| x1 | 1 | 0.4 | -0.2 | 0 | 0.2 | 4 |
| s2 | 0 | 0.2 | 0.4 | 1 | -0.4 | 2 |

| Term | Value | Meaning |
| --- | --- | --- |
| **(Nuts)x1** | **4 kg** | Use 4 kg of nuts in the mix. |
| **(Oats)x2** | **0 kg** | Do not use oatmeal; it's not cost-effective for these targets. |
| **(Slack Fat)s2** | **2 units** | As fat limit was 10. You only used 8. have 2 units of "buffer." |
| **(Surplus Protein)s1** | **0 units** | Provided exactly 20 units of protein. No waste. |
| **(Min Cost)Z** | **$32.00** | This is the cheapest way to make the bar. |

If you try to use more Oatmeal ($x_2$) to save money (since it's only $5/kg), you would have to reduce Nuts ($x_1$). Because Nuts are much higher in protein, you would eventually break the "At least 20 Protein" rule. The Simplex math found the perfect balance at 4 kg of Nuts.

A factory produces two types of energy drinks: **Power ($x_1$)** and **Focus ($x_2$)**.

- **Goal:** Maximize Profit $Z = 4x_1 + 3x_2$
- **Constraint 1 (Machine Time):** $2x_1 + 3x_2 \leq 12$ (Must not exceed 12 hours)
- **Constraint 2 (Exact Orders):** $x_1 + x_2 = 5$ (Customer requires exactly 5 units total)

**Step 1: Standard Form**

- **Eq 1 ($\leq$):** $2x_1 + 3x_2 + s_1 = 12$ (We add a **Slack** variable)
- **Eq 2 ($=$):** $x_1 + x_2 + A_1 = 5$ (Since it's an equality, we only add an **Artificial** variable to get a starting basic solution)
- **Objective:** Maximize $Z = 4x_1 + 3x_2 - MA_1$ *(Note: For Maximization, we **subtract** $M$ because $A_1$ is a penalty we want to get rid of.)*

| **Basic** | **x1** | **x2** | **s1** | **A1** | **RHS** |
|---|---|---|---|---|---|
| Z | -M-4 | -M-3 | 0 | 0 | -5M |
| S1 | 2 | 3 | 1 | 0 | 12 |
| A1 | **[1]** | 1 | 0 | 1 | 5 |

- **Entering:** $x_1$
- **Ratio Test:** $12/2 = 6$ vs $5/1 = \mathbf{5}$.
- **Pivot Element:** **1** (in the $A_1$ row).

1. Minimize Cost $Z = 5x_1 + 4x_2$

**Subject to:**

1. $3x_1 + 2x_2 \geq 12$ (Minimum Strength)
2. $x_1 + 2x_2 \geq 8$ (Minimum Weight)
3. $x_1, x_2 \geq 0$

2.

- **Minimize Cost:** $Z = 10x_1 + 15x_2$
- **Constraint 1 (Reactive Balance):** $2x_1 + 1x_2 \geq 8$
- **Constraint 2 (Total Volume):** $x_1 + x_2 = 6$
- **Non-negativity:** $x_1, x_2 \geq 0$

1. **For $\geq$:** Subtract a Surplus ($s_1$) and add an Artificial ($A_1$).
    - $2x_1 + 1x_2 - s_1 + A_1 = 8$
2. **For $=$:** Add an Artificial ($A_2$).
    - $x_1 + x_2 + A_2 = 6$
3. **Objective Function:** Add the Big M penalties for both artificial variables.
    - Minimize $Z = 10x_1 + 15x_2 + MA_1 + MA_2$

# Sensitivity Analysis

## Range of Optimality

This refers to the range over which an **objective function coefficient** (like the profit per unit) can change without changing the **optimal values of the decision variables**

**Example:** selling lemonade for $2.00. The Range of Optimality might be $1.50 to $3.00. If we raise the price to $2.50, we should still make the exact same number of cups; you'll just make more money. If it hits $3.01, you might need to switch your strategy entirely.

## Shadow Prices (Dual Values)

A **Shadow Price** tells you how much your total objective value (profit) would improve if you had **one additional unit** of a resource.

**Example:** You are limited by 100 hours of labor. If the shadow price for labor is $15, then getting 101 hours of labor would increase your total profit by $15.

**Note**- This is only valid within the **Range of Feasibility**. If you buy *too* much labor, another resource the shadow price will change.

# Degeneracy

Degeneracy occurs in the Simplex method when one or more **basic variables are zero**.